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Generative transfer optimization for more efficient turbine blade design

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ABSTRACT

In order to accelerate the turbine blade design, a generative transfer optimization (GTO) framework is proposed with generative adversarial net (GAN). First, a gradient-based reparameterization strategy combined with an sample-weighted training scheme is proposed for GAN in order to enable transfer optimization. Wherein, the gradient-based reparameterization strategy is used to affine solutions of related source task (a.k.a. source samples) that represented by coordinate points to the latent design space of the optimization problem at hand. The sample-weighted training scheme helps to learn the distinct features of source samples and thus increasing the proportion of solutions with better performance in the target design space. After that, a multi-fidelity surrogate based transfer optimization strategy is used, which augments the limited target samples by source samples to attain the optimal solutions more quickly. Using the proposed GTO framework, optimization is conducted for a turbine blade with high loadings, which is also compared with the conventional methods. It shows that the proposed GTO can finish the optimization task more quickly and attain much better solutions. When the goal is to achieve solutions with similar performance, the computational cost of GTO can be reduced by 50% and even more.

1. Introduction

The turbine blade design is an expensive black-box problem, the method combined with optimization techniques provides an efficient way to attain high-performance turbine blade design [1–4]. However, with the increasing refinement of turbine blades, researchers need to use more sophisticated and more expensive CFD methods to carry out blade design under higher degrees of freedom; at the same time, the allowable time for turbine design continues to shorten. Under these circumstances, it is urgent to develop more efficient design optimization methods to achieve ever more refined turbine designs without increasing or even decreasing the optimization costs.

The optimization of turbine blade consists of three basic elements, i.e., optimization algorithm, parameterization method, and performance evaluation method. Among them, optimization algorithm controls the optimization progress; parameterization method determines the upper limit of the performance of the attained final solutions, while the number of variables used for parameterization determines the cost of optimization, particularly when the global optimization algorithms such as efficient global optimization (EGO) algorithms [5,6], genetic algorithms are used [7,8]. For these global optimization algorithms, the number of function evaluations required increases exponentially with the number of design variables [9,10].

In order to reduce the optimization cost, the most common idea is to minimize the number of variables that involved in the optimization process. To achieve it, the first method is to use sensitivity analysis methods to screening out the most significant variables of design space that parameterized by classical parameterization methods [11] such as NURBS, FFD, and then only the significant variables in the parameter space are optimized [12,13]. The second type of method is to use Proper Orthogonal Decomposition (POD) and etc. to reduce the variable dimension of the blade profile that initially represented by hundreds of scatter points [14,15]. Along with the rapid development of AI technology, the generative model based blade parameterization method proposed [16–19]. The basic idea is to use GAN and other generative models to extract features from the scatter points of blade profile, and then use several or a dozen so-called latent space variables to parameterize the turbine blades. For example, Prof. Fuge and this group members [20] propose Bezier-GAN, which only uses three variables to parameterize the aerodynamic shape of airfoils, and consequently the computational cost of related optimization is only 1/10 or even less when comparing to those methods that using NURBS, FFD and etc. for the blade parameterization. However, it should be noted that, whether it is the traditional dimensionality reduction methods or the latest generative model-based parameterization methods, the corresponding optimization methods face the trade-off between optimization cost and optimization performance:

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i.e., a design with more design variables often results in a better performance; while a reduction in the number of design variables reduces the cost of the optimization, but the performance of the optimized solution will also be degraded. How to break through the above trade-off and obtain better design performance with lower optimization cost without reducing the number of design variables is a bottleneck that needs to be resolved in the research of design optimization methods.

The recently proposed transfer learning (TL) [21] provides a novel way to alleviate the trade-off between cost and performance as noted above. Inspired by the fact that designers can often take full advantage of past experience to better solve the problem at hand, the key idea of TL is to gain knowledge from related tasks to help solve the target task more efficiently. Prof. Ong and his group members introduce the concept of transfer learning in the study of optimization algorithms and proposed the concept of transfer optimization (TO) [22,23], which has drawn wide attentions [24–26]. Similar to “copying homework”, the key issue in TO research is how to make use of the solutions of completed or ongoing related tasks to help the algorithm reach the vicinity of the global optimal solution of the target task more quickly.

It is also important to note that most of previous studies focus on the knowledge transfer strategies study of optimization algorithms. However, for the turbine design, although there exists a large number of aerodynamic optimization tasks with similar operating conditions and blade profiles, it is often difficult for researchers to carry out turbine blade optimization using a unified parameterization method. In other words, when applying transfer optimization for turbine design, one issue needs to be resolved, i.e., how to parameterize the solutions of the related completed/ongoing tasks into the target design space to speed up the optimization progress [27].

To address the above issue, shared latent space is built for the target optimization problem and related source tasks using GAN, wherein a gradient-based strategy is used to encode the optimized solutions of source task into the shared latent space, and a multi-fidelity surrogate transfer optimization strategy is presented to accelerate the progress of the target optimization task.

The remainder of this paper is organized as follows: The preliminaries of related topics are presented in Section 2. Then, the proposed transfer optimization framework is illustrated in Section 3. After that, a turbine transfer optimization is conducted in Section 4, and conclusions are drawn in Section 5.

2. Preliminaries

2.1. Classical design optimization flowchat of turbine blade

The blade design optimization is conducted by formulating the design task as the following numerical optimization problem:

$$\min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}) \quad (1)$$

where, $\mathbf{x}$ is an sample in the design space $\mathcal{X}$, and $f(\cdot)$ is the objective function of the design task. The standard optimization procedure of turbine blades is carried out in two stages, as shown in Fig. 1. In the first stage, the turbine blade, which is initially represented by hundreds of coordinate scatters, is parameterized with dozens of variables. Usually, the design space of coordinate points is labeled as $\mathcal{X}$, while the design space of parameterized variables is denoted by $\mathcal{Z}$. The methods such as FFD are often employed for blade parameterization. Then, in the second stage, by combining the parameterization method with an optimization algorithm, an optimization of the turbine blades is conducted.

Since it is very expensive to evaluate sample performance using CFD softwares, surrogate-based optimization algorithms such as efficient global optimization (EGO), also known as Bayesian optimization (BO) are often used. The flowchart of EGO is as follows. First, a surrogate is built using the Gaussian process (GP) to approximate the relation between the input variables and the objective function. Second, an infill criterion such as expected improvement (EI) is employed to decide

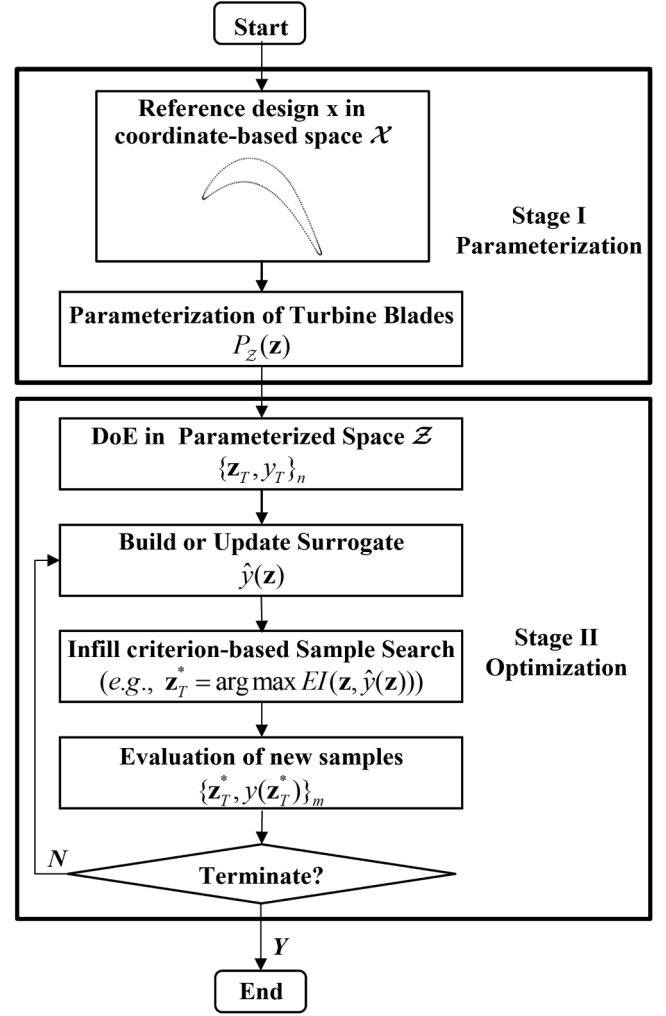


Fig. 1. Standard flowchart of turbine blade optimization procedure.

the next candidate solution to evaluate. Third, the training set and the surrogate are updated for the next optimization cycle.

As a key element of the EGO algorithm, GP [28] is a popular surrogate technique that treats the function predictor $Y(\mathbf{z})$ as a Gaussian random variable:

$$Y(\mathbf{z}) \sim \text{GP}(m(\mathbf{z}), k(\mathbf{z}, \mathbf{z}')) \quad (2)$$

where $m(\mathbf{z})$ is the mean value of the objective function to predict, and $k(\mathbf{z}, \mathbf{z}')$ is the spatial correlation function between the variable vector $\mathbf{z}$ and $\mathbf{z}'$. Usually, $m(\mathbf{z}) \equiv 0$ is imposed, and the radial basis covariance function is employed for $k(\mathbf{z}, \mathbf{z}')$. For standard GP with single-fidelity samples, the prediction of the objective function ($\hat{y}$) and the related prediction uncertainty (s^2) at an sample point $\mathbf{z}$ are as follows:

$$\begin{aligned} \hat{y}(\mathbf{z}) &= \mathbf{k}(\mathbf{z}, \mathbf{Z})(\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y} \\ s^2(\hat{y}(\mathbf{z})) &= \mathbf{k}(\mathbf{z}, \mathbf{z}) - \mathbf{k}(\mathbf{z}, \mathbf{Z})(\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{k}(\mathbf{Z}, \mathbf{z}) + \sigma_n^2 \end{aligned} \quad (3)$$

where, σ_n^2 is the Gaussian noise, $\mathbf{K}$ is the covariance matrix of the training samples, and $\mathbf{k}$ is the vector that measures the spatial correlations between $\mathbf{z}$ and $\mathbf{Z}$.

In addition to the surrogate, the infill criterion helps the EGO algorithm to decide which sample is promising to be queried in the next optimization cycle. Among various infill criteria, the expected improve-

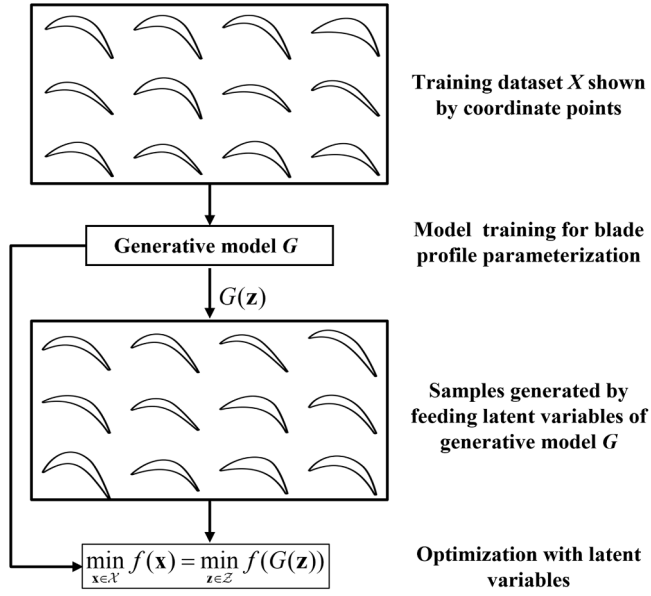


Fig. 2. Flowchart of generative model based optimization.

ment (EI) [5] is most popular, which is written as follows:

$$EI(\mathbf{z}) = (f_{\min}^{(t)} - \hat{f}(\mathbf{z}))\Phi(u) + s(\mathbf{z})\phi(u) \quad (4)$$

$$u = (f_{\min}^{(t)} - \hat{f}(\mathbf{z}))/s(\mathbf{z})$$

where, $f_{\min}^{(t)}$ is the current best solution, and $\Phi(\cdot)$ and $\phi(\cdot)$ are the standard Gaussian probability distribution and density functions, respectively. In each iteration, the next candidate solution to evaluate is chosen with the following equation:

$$\mathbf{z}^* = \arg \max_{\mathbf{z}} EI(\mathbf{z}) \quad (5)$$

2.2. Generative model based shape optimization

The parameterization method and optimization algorithm are the two key elements for turbine design optimization. In previous studies, the conventional parameterization methods such as NURBS and FFD are frequently used, which parameterizing the aerodynamic shapes with a single reference design. Recently, with the advent of generative models such as GAN, the data-driven generative model-based design optimization method (GMO) has drawn wide attentions, which is also known as latent space optimization (LSO) [29].

The general flowchart of GMO is shown in Fig. 2. First, a training dataset X is collected that representing aerodynamic profiles by coordinate-based scatter points. Second, with the dataset X , the generative model G such as GAN is used to learn the latent space $\mathcal{Z}$ that controlled by a few variables.

Then, by feeding a variable vector $\mathbf{z}$ into the generative model, i.e., $G(\mathbf{z})$, various aerodynamic shapes can be attained. Then, the optimization can be conducted as follows:

$$\min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}) = \min_{\mathbf{z} \in \mathcal{Z}} f(G(\mathbf{z})) \quad (6)$$

Compared to conventional parameterization methods such as NURBS and FFD, the generative model has shown the capability to learn a design space using fewer variables but presenting better diversity of samples, and therefore solves the problem more quickly with better solutions [16, 20, 30, 31].

2.3. Transfer optimization

The key concept of transfer optimization is that, the target problem T can be solved more efficiently with the help of related completed tasks

$S_1, \dots, S_m$, which is formulated as follows:

$$T = \min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}) \rightarrow \min\{T | S_1, \dots, S_m, K(t)\} \quad (7)$$

$$\text{where, } S_m = \min_{\mathbf{x}_m \in \mathcal{X}_m} f_m(\mathbf{x}_m)$$

where, $\mathbf{x}$ is a sample of the target problem, and $\mathcal{X}$ is the design space of the optimization problem at hand, and $\mathcal{X}_m$ denotes the design space of completed tasks. In the meantime, $K(t)$ is the strategy that draws useful knowledge from the related tasks to facilitate the optimization of the target optimization problem. The form of $K(t)$ can be very flexible. For example, some studies directly inject the solutions of completed tasks to speed up the progress of target task [23]. Alternatively, some other studies extract useful knowledge from the samples of related source tasks to build more accurate approximations with multi-fidelity surrogate [27, 32].

Also note that, there is a precondition in $K(t)$. That is, the solutions of related source task should be easily affined to the parameterized design space of the target problem, which is rarely investigated in previous studies but will be resolved in this paper.

3. Generative transfer optimization

As noted in the section on preliminaries, when applying transfer optimization to speed up the design progress, the following issues have to be resolved, i.e. (1) how to build a shared latent space for both target problem and related source tasks; (2) how to affine the solutions of the source tasks to the shared latent space; (3) how to make use of the solutions of the source tasks to speed up the optimization progress. These issues are addressed in this section. After that, a framework of generative transfer optimization for turbine blade design is proposed.

3.1. Parameterization with sample-weighted GAN

Recalling the scenario in real situations, as the CFD-based performance evaluation is conducted by taking the blade profiles represented by hundreds of scatter points as the input, the blade profile of optimal solutions denoted by coordinates of scatter points will be stored, while the parameterization methods used in blade optimization may not be stored. Then, when conducting optimization for similar turbine blades, how to affine the solutions of completed tasks to the design space of the problem to solve may not be an trivial work.

When we combine the source and target samples together and representing them in the form of coordinate-based scatter points, a shared latent design space can be learned by using generative deep learning models such as GAN. Thereafter, the blade optimization can be conducted in the latent space. Along this idea, a sample-weighted training scheme is proposed for the GAN training.

The GAN [33] is made up of two components, i.e., the generator and discriminator, as shown in Fig. 3. The following loss function is used for the training of GAN:

$$L(\mathbf{x}) = \min_G \max_D V(D, G) + \lambda L_p(\mathbf{x}) \quad (8)$$

where, $\min_G \max_D V(D, G)$ is the adversarial loss of standard GAN model, which is formulated as follows:

$$\min_G \max_D V(D, G) = E_{\mathbf{x} \sim P_{data}} [\log D(\mathbf{x})] + E_{\mathbf{z} \sim P_z} [\log (1 - D(G(\mathbf{z})))] \quad (9)$$

where, $D(\mathbf{x})$ is the probability that the sample coming from the real dataset, while $D(G(\mathbf{z}))$ is the probability that the sample is synthesized by the generator.

After training, the design space of blade profiles is implicitly encoded. By varying the latent vector $\mathbf{z}$, various blade profiles can be generated. To avoids geometric flaws such as crossed curves and abrupt corners, a regularization term L_p is added in Eq. (9) to generate smooth profiles that can meet the design requirements of turbine blades. Since

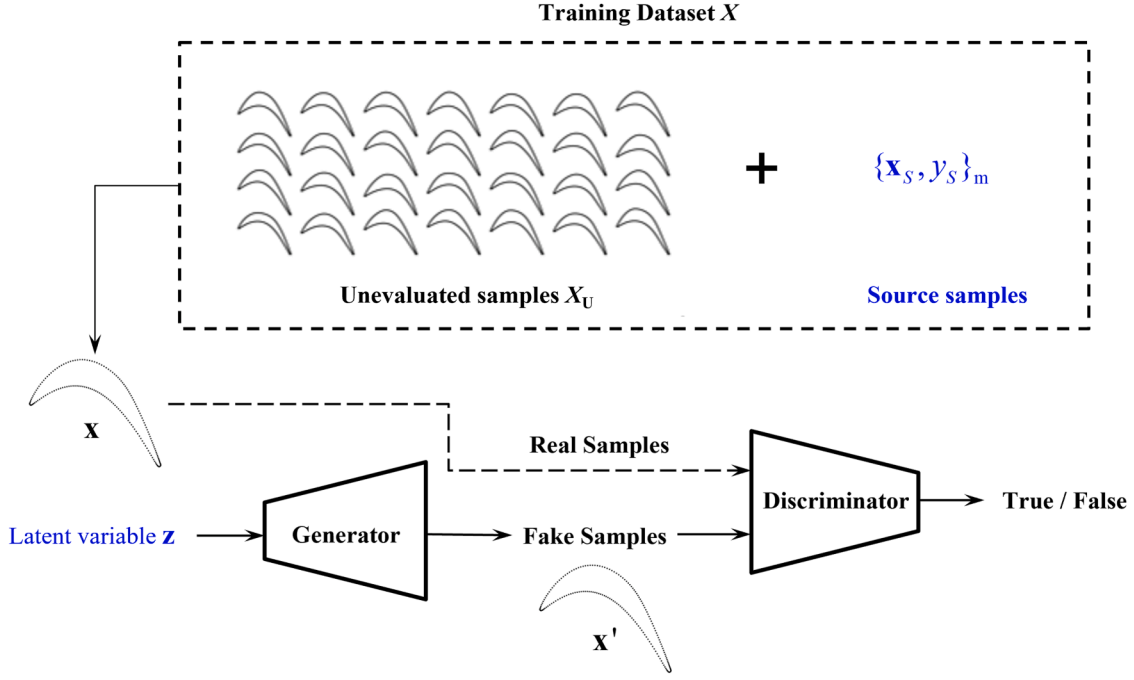


Fig. 3. Shared latent design space trained by sample-weighted GAN.

the term L_p may vary case to case, which will be illustrated with more details in Section 4.

On one hand, GAN training needs thousands of samples; on the other hand, the samples that can be gathered from related source tasks is usually in the amount of dozens at most. Therefore, the number of the source samples (labeled as $\{\mathbf{x}_S, \mathbf{y}_S\}_m$ in Fig. 3) is far less than the unlabeled samples (labeled as X_U). Then, if we combine $\{\mathbf{x}_S, \mathbf{y}_S\}_m$ with X_U directly for the GAN training, the features of the source samples will be lost. An sample-weighted training scheme is proposed to resolve the above problem with the following loss function:

$$L_w = \sum_{i=1}^{n^{(S)}} L(\mathbf{x}_i^{(S)}) + \sum_{j=1}^{n^{(T)}} L(\mathbf{x}_j^{(T)}) \quad (10)$$

where, $L(\cdot)$ is drawn from Eq. (8), and $\mathbf{x}_i^{(S)}$ and $\mathbf{x}_j^{(T)}$ are the samples that randomly selected from $\{\mathbf{x}_S, \mathbf{y}_S\}_m$ and X_U , respectively. The number of samples that drawn from X_U or $\{\mathbf{x}_S, \mathbf{y}_S\}_m$ in a batch can be changed. For example, in order to learn the features of the source samples, we can impose $n^{(S)}/n^{(T)} = 1.0$, i.e., half of the samples are come from $\{\mathbf{x}_S, \mathbf{y}_S\}_m$ in each training epoch.

3.2. Gradient-based reparameterization of source samples

Here, we define the reparameterization as encoding the source samples to the latent space of the target optimization task, which is a prerequisite to apply transfer optimization for turbine blade design. However, such reparameterization cannot be that easy to be implemented using GAN, since GAN does not support the inverse inference from the coordinate-represented sample $\mathbf{x}$ to the parameterized variable vector $\mathbf{z}$.

Inspired by the back-propagation algorithm of neural network training, the reparameterization of source samples can be interpreted as an optimization problem with the following equation:

$$\mathbf{z}^* = \arg \min_{\mathbf{z}} \|G(\mathbf{z}) - \mathbf{x}\|_2^2 \quad (11)$$

The gradient-based algorithm that uses the automation differentiation operator of PyTorch is used for the above minimization problem. The iteration in each step is written as follows.

$$\mathbf{z}^{(n+1)} = \mathbf{z}^{(n)} - \eta \cdot \frac{\partial \|G(\mathbf{z}^{(n)}) - \mathbf{x}\|_2^2}{\partial \mathbf{z}^{(n)}} \quad (12)$$

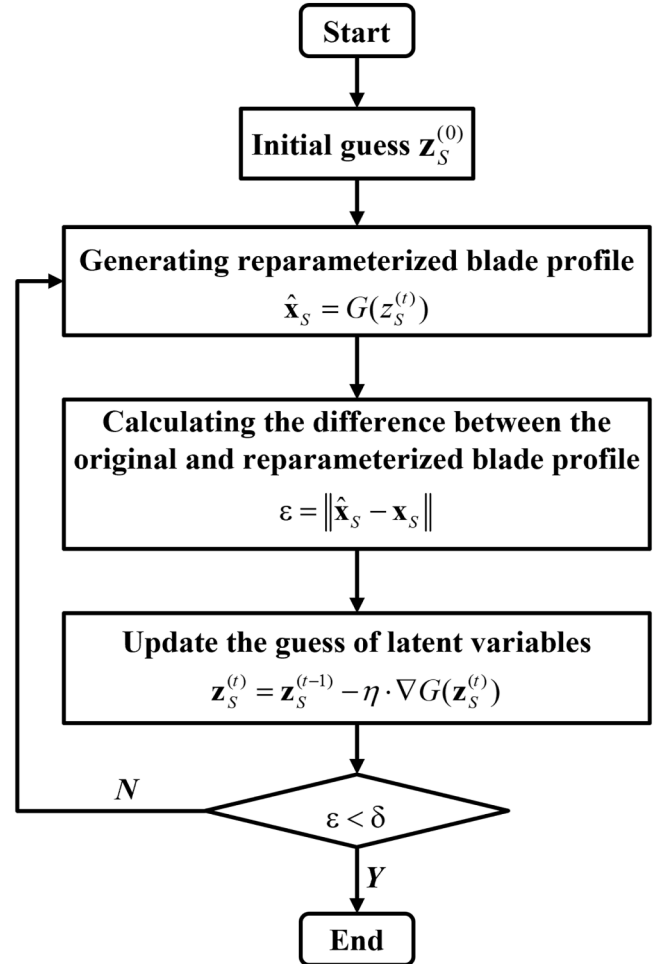


Fig. 4. Flowchart of gradient-based reparameterization of source samples.

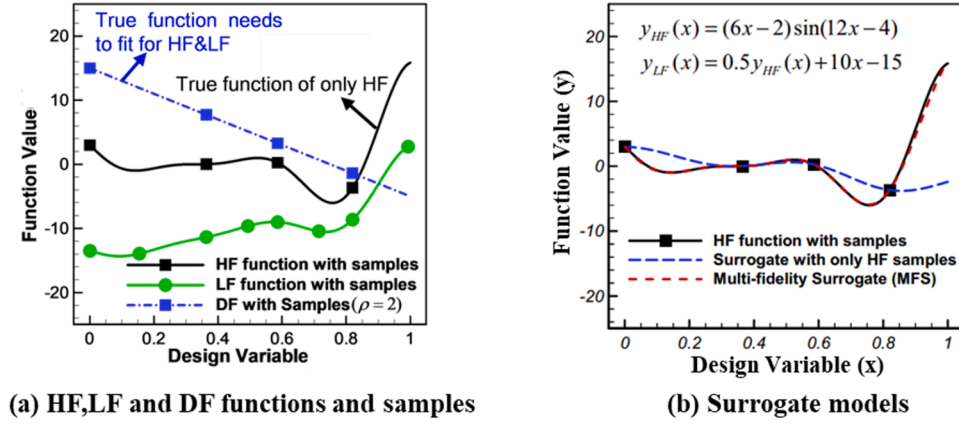


Fig. 5. Schematic of multi-fidelity surrogate-based knowledge transfer by taking the source samples as low-fidelity(LF) samples [35].

Where, η is the learning rate. Accordingly, the iteration flowchart can be found in Fig. 4, and the effectiveness of such reparameterization will be tested in Section 4.

3.3. An ensemble-weighted multi-fidelity surrogate for transfer optimization

After mapping the source samples to the target design space, how to make use of these source samples is another problem to be solved for transfer optimization. Since the objective functions of the related source task are similar to, but different from that of the target problem, so the samples of source tasks is treated as low-fidelity (LF) samples and the target samples as high-fidelity (HF) samples to build a multi-fidelity surrogate (MFS) [34], which is believed to be more accurate than the surrogate approximation built by HF samples alone. Thereby, the MFS-based optimization, which draws information from the source samples, is expected to achieve the optimal solutions more quickly with fewer cost.

The mathematical expression of MFS can be expressed as follows:

$$\hat{y}_{MFS}(\mathbf{x}) = \rho \cdot \hat{y}_{LF}(\mathbf{x}) + d(\mathbf{x}) \quad (13)$$

where, ρ and $d(\cdot)$ are the scaling factor and the discrepancy function (DF) between the high-fidelity (HF) and low-fidelity (LF) models, respectively. The effectiveness of the multi-fidelity surrogate can be illustrated through the one-dimensional toy problem shown in Fig. 5. As can be seen from Fig. 5(a), the overall trends of the HF and LF functions in the design space are similar. In the meantime, the HF curve can be hardly fit by the four HF samples alone (shown by squared dots), as seen in Fig. 5(b). However, when combining the small number of HF samples with relatively large number of LF samples to build the surrogate, the fitting of original complex HF function can be changed to the fitting of the discrepancy function shown by the dash-dotted line, which can be much easier. Consequently, the true curve of HF function is observed to be perfectly fitted by MFS (denoted by the short dashed line) in Fig. 5(b). To summarize, the involvement of low-fidelity samples, which have similar trend with the target HF objective function, can help to reduce the fitting complexity of the original HF function and thus providing more accurate guidance for the following optimization search.

Also note that, the single-fidelity surrogate (SFS) built with HF samples alone may be more accurate than that of MFS when the function trend of HF and LF models are not well correlated. Alternatively, with the increase of HF samples, the SFS can also become more accurate than that of MFS. Then, negative transfer may happen, i.e., the optimization with HF samples alone may achieve better final solutions than that combining HF and LF samples. To address the above issue, an ensemble-weighted surrogate that combining SFS and MFS is used for transfer optimization, labeled as EMFS [36]. The function prediction and related prediction uncertainty of EMFS are shown as

follows:

$$\begin{aligned} \hat{y}_{EMFS}(\mathbf{x}) &= w_G(\mathbf{x}) \cdot \hat{y}_{MFS}(\mathbf{x}) + w_L(\mathbf{x}) \cdot \hat{y}_{SFS}(\mathbf{x}) \\ s_{EMFS}^2(\mathbf{x}) &= w_G(\mathbf{x}) \cdot s_{MFS}^2(\mathbf{x}) + w_L(\mathbf{x}) \cdot s_{SFS}^2(\mathbf{x}) \end{aligned} \quad (14)$$

where, w_G and w_L are the weights of SFS and MFS built using Gaussian process. More specifically, w_G and w_L are calculated using the following equations:

$$\begin{aligned} w(\mathbf{x}) &= s_L^2(\mathbf{x}) / (s_L^2(\mathbf{x}) + s_G^2(\mathbf{x})) \\ w_L(\mathbf{x}) &= \text{sigmoid}(w(\mathbf{x})) = 1 / (1 + e^{-(m \cdot w(\mathbf{x}) - b)}) \\ w_G(\mathbf{x}) &= 1 - w_L(\mathbf{x}) \end{aligned} \quad (15)$$

where, s_G^2 is the prediction uncertainty measured by the single-fidelity Gaussian process trained with all HF samples, and s_L^2 is the prediction uncertainty measured by the Gaussian process trained with local HF samples in the area where samples are densely distributed, and such area is detected by the DBSCAN algorithm [37]. More intuitively, the schematic of EMFS can be illustrated through the toy problem shown in Fig. 6. The DBSCAN algorithm is first used to detect the areas where HF samples are densely distributed. Then, a local SFS is built with the clustered HF samples in that area. In the meantime, an MFS is built with both LF and HF samples. And the weights of SFS and MFS for the EMFS are calculated using Eq. (15). As can be seen in Figure, the MFS can capture the overall trend of the true function, but the local SFS has much better accuracy in the region where HF samples are relatively more densely distributed. Then, by combining MFS and the local SFS, the EMFS is shown to approximate the true function most accurately. More details of EMFS can refer to [36].

After building the multi-fidelity surrogate that combining HF and LF samples, the EI function shown in Eq. (4) is reformulated as follows:

$$\begin{aligned} EI_M(\mathbf{z}) &= (f_{\min}^{(t)} - \hat{y}_M(\mathbf{z}))\Phi(u_M) + \sigma_M(\mathbf{z})\phi(u_M) \\ u_M &= (f_{\min}^{(t)} - \hat{y}_M(\mathbf{z})) / \sigma_M(\mathbf{z}) \end{aligned} \quad (16)$$

where, the subscript M is used to differentiate it from the surrogate that built using HF samples alone. Similar to Eq. (4), $\Phi(\cdot)$ and $\phi(\cdot)$ are the standard Gaussian distribution and density function, respectively, and $f_{\min}^{(t)}$ is the current best solution.

3.4. Framework of generative transfer optimization

By combining the sample-weighted GAN, gradient-based reparameterization method of source samples with multi-fidelity surrogate based optimization strategy, a framework of generative transfer optimization for turbine blade design is proposed, as shown in Fig. 7.

Similar to the optimization procedure shown in Fig. 1, it consists of two stages, i.e., parameterization and optimization. However differently, a shared design space that including the source samples is built using GAN. More importantly, after reparameterizing the source samples

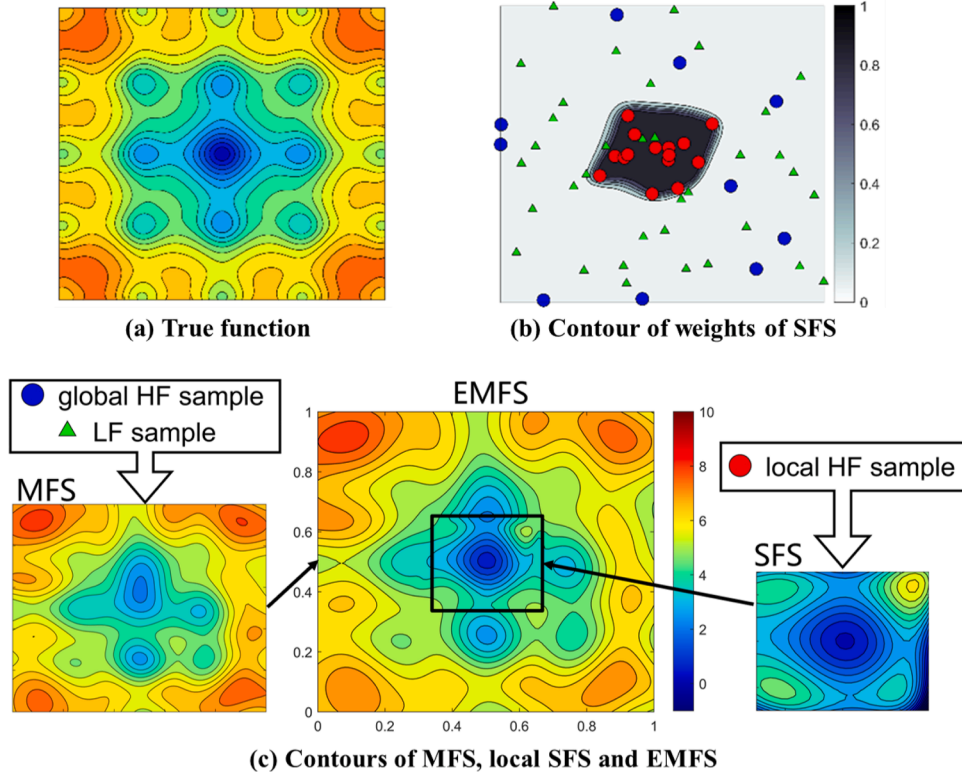


Fig. 6. Illustration of EMFS with two-dimensional Ackley function [36].

that originally represented by coordinate-based scatter points, a multi-fidelity surrogate (i.e. $\hat{y}_M$) is built to make full use of source samples. With the above, the generative transfer optimization for turbine blades is carried out as follows.

First, the dataset X is collected that includes both samples of source tasks (i.e., $\{x_S, y_S\}_m$) and unevaluated samples (i.e., X_U), where $\{x_S\}_m$ and X_U are represented by hundreds of coordinate-based points, and X_U are often generated using Latin hypercube sampling technique [38] by referring to the reference design of the target problem.

Second, to learn the shared parameterized design space $\mathcal{Z}$ for the target optimization problem, a sample-weighted GAN is trained with X .

Third, using the gradient-based reparameterization method illustrated in Fig. 4, the source samples $\{x_S, y_S\}_m$ are parameterized into the target design space, denoted by $\{\hat{z}_S, y_S\}_m$.

Fourth, using the sample generator $G(z)$ with Latin hypercube sampling technique, the target samples are generated, labeled $\{\hat{z}_T, y_T\}_n$.

Fifth, combining $\{\hat{z}_S, y_S\}_m$ with $\{\hat{z}_T, y_T\}_n$, a multi-fidelity surrogate is built using EMFS shown in Eq. (14), and an iterative optimization will be carried out by maximizing the EI equation shown in Eq. (16). The iterative optimization will be terminated once the maximum allowable number of optimization cycles was met.

4. Experimental study: Aerodynamic optimization of a highly loaded turbine blade

4.1. Problem description

In order to mimic the scenario of transfer optimization in real-world problems, two design tasks with similar working conditions are constructed using the GE-E3 low-pressure turbine (LPT) blade as the reference. Task 1 is to optimize the blade profile for the original GE-E3 LPT blade at the design point, which is assumed to have been finished and will be used as a source task, and Task 2 is to explore blades of high loadings in response to boosting the thrust-to-weight ratio of the aero-engine, and the highly-loaded working condition is built by increasing

Table 1

Settings of the original and highly turbine blade.

Specifications	Original	High-loaded
Inlet flow angle / °	49.3	49.3
Inlet total pressure / Pa	319339.2	319339.2
Inlet total temperature / K	1161	1161
Outlet static pressure / Pa	245,618	245,618
Axial chord length / mm	26.6	26.6
Pitch / mm	25.006	30.0072
Outlet flow angle / °	-64.19	-60.86

the pitch of original GE-E3 LPT blade. The specifications of the referenced design of the two design tasks are shown in Table 1, and the mathematical expressions of the optimization model for Task 1 and Task 2 are as follows:

$$S = \min \varepsilon_p^O(\mathbf{x}),$$

$$\text{s.t. } 0.98 \cdot \dot{m}_{ref}^O \leq \dot{m}(\mathbf{x}) \leq 1.02 \cdot \dot{m}_{ref}^O, \quad (17)$$

$$r_{\min}(\mathbf{x}) \geq r_{\min,ref}$$

$$T = \min \varepsilon_p^H(\mathbf{x}),$$

$$\text{s.t. } 0.98 \cdot \dot{m}_{ref}^H \leq \dot{m}(\mathbf{x}) \leq 1.02 \cdot \dot{m}_{ref}^H, \quad (18)$$

$$r_{\min}(\mathbf{x}) \geq r_{\min,ref}$$

where, ε_p is the total pressure loss at cascade outlet, and S is short for the source task, and T is short for the target task. The superscript O in Eq. (16) means it refers to the original GE-E³ blade, while H in Eq. (17) means it refers to the highly loaded LPT blade. The minimum radius of the GE-E³ LPT blade (denoted by $r_{\min,ref}$) is 0.2mm, which cannot be further reduced to maintain the blade mechanical strength.

Before detailed discussion of the formulation of the transfer optimization task, the numerical model for the performance evaluation of highly-loaded LPT blade is verified, as shown in Fig. 9. Specifi-

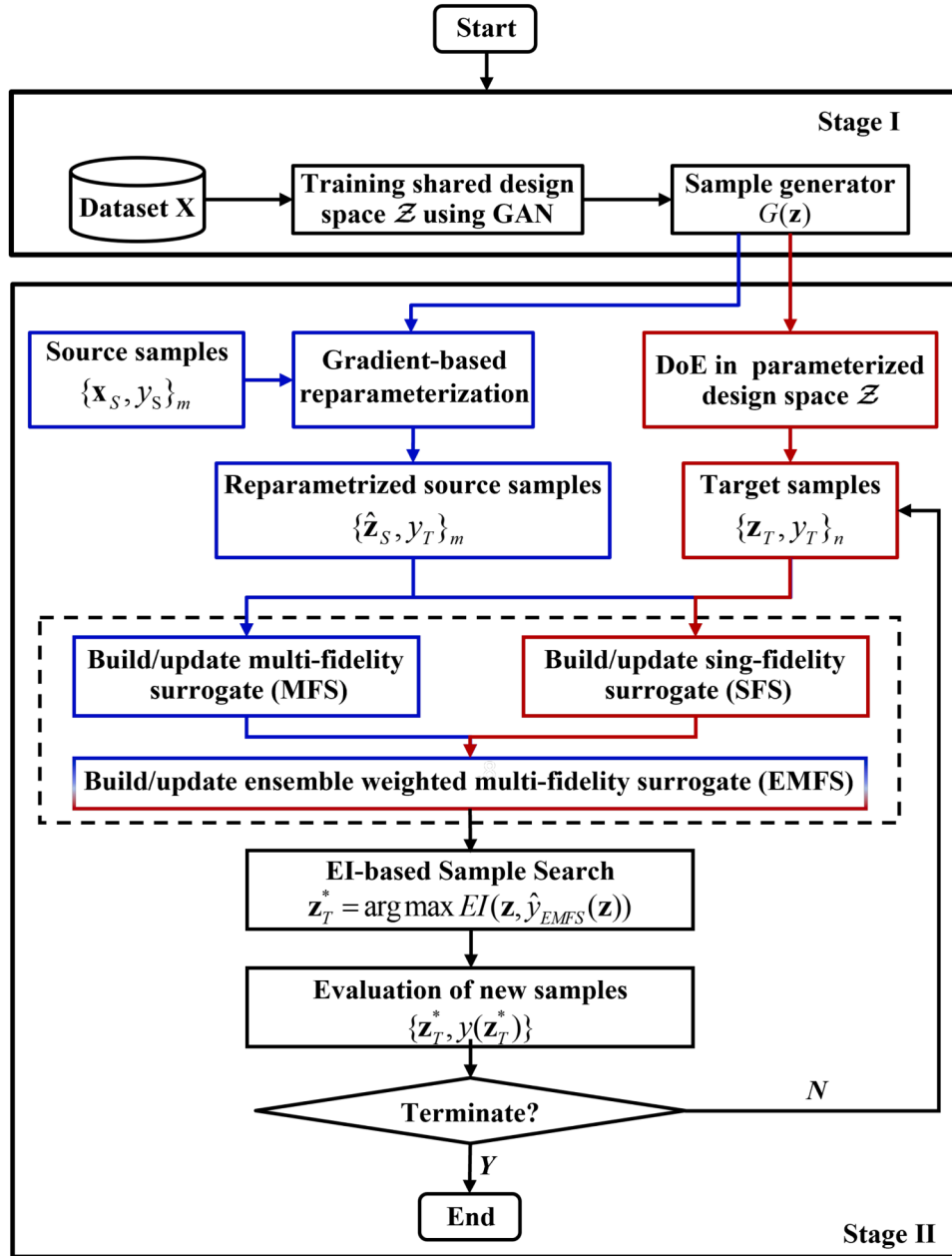


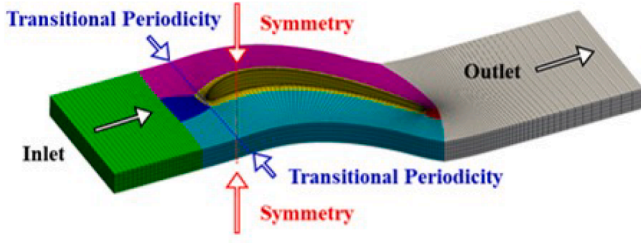
Fig. 7. Flowchart of the generative transfer optimization for turbine blade design.

cally, the Reynolds-averaged Navier-Stokes (RANS) solver is used. The shear-stress transport (SST) turbulence model coupled with the gamma theta transition is adopted. In the meantime, for the linear cascade, the boundary condition of the hub and shroud is set as symmetry, and the transitional periodicity is used. After grid independence verification, the number of grid cells is set to 492700, and the same grid template is used for the following optimization process.

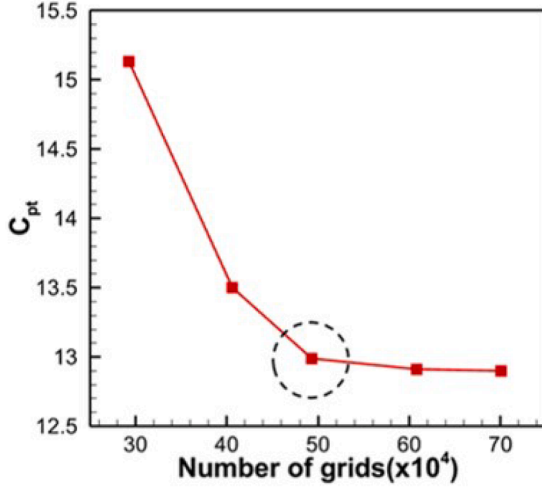
The transfer optimization scenario of the highly-loaded turbine can be intuitively described with Fig. 9. That is, the referenced design profile and the solutions of the related source task are given. The issue is how to make use of these resources to solve the target task efficiently. It should also be noted that, the optimized solutions of the source task may come from other researchers but not our team. In other words, the prescribed parameterization method that used for the source tasks is unavailable, and the solutions of source task are shown by hundreds

of coordinate points. Then, how to map the source task samples to the parameterized design space of the target task is the first key problem to be solved for transfer optimization. This may explain why transfer optimization is rarely investigated in previous studies for the turbine design.

In general, three design frameworks can be used for the aerodynamic optimization of highly loaded turbine blades, as shown in Fig. 8. The first type of design optimization framework is the one that commonly used by current researchers, i.e., the classical parameterization method (e.g., NURBS and etc.) is first employed to parameterize the blade contour, and then design optimization is conducted in the parameterized design space. Under this classical framework, the FFD is selected to parameterize the turbine blade, and the well-known EGO algorithm shown in Section 2.1 is used as the optimizer to conduct optimization. The above method is denoted by FFD-EGO in the following paragraphs. The expres-



(a) Numerical model



(b) Grid independence verification

Fig. 8. Numerical model and grid independence verification for the highly-loaded turbine blade.

sion for the optimization over parameterized design space of FFD-EGO is as follows:

$$\begin{aligned} T &= \min \varepsilon_p^H(\mathbf{P}(\mathbf{z})), \\ \text{s.t. } 0.98 \cdot \dot{m}_{ref}^H &\leq \dot{m}(\mathbf{z}) \leq 1.02 \cdot \dot{m}_{ref}^H, \\ r_{\min}(\mathbf{z}) &\geq r_{\min,ref} \end{aligned} \quad (19)$$

where, $\mathbf{z}$ is the variable of parameterized design space using FFD. More setting details of FFD-EGO will be illustrated in Section 4.3.

The second type of design optimization framework is the generative model based optimization (GMO), which has drawn wide attentions recently and is also known as latent space optimization. As illustrated in Section 2.2, the GMO is conducted as follows. First, a dataset of blade profiles is built; Second, a latent design space is built with the dataset; Third, optimization is conducted in the latent design space. Under GMO framework, a modified version of Pritchard's geometric parameterization method [39] is used to build the unlabeled dataset X_U by taking the highly loaded turbine blade shown in Table 1 as the reference design. Moreover, the source samples (i.e., $\{\mathbf{x}_S, \mathbf{y}_S\}_m$) is added to the training dataset. Similar to FFD-EGO, the EGO algorithm is used as the optimizer to conduct optimization. The above design optimization method is denoted by GAN-EGO in the following paragraphs. The expression of the optimization over parameterized design space of GAN-EGO is as follows:

$$\begin{aligned} T &= \min \varepsilon_p^H(\mathbf{G}(\mathbf{z})), \\ \text{s.t. } 0.98 \cdot \dot{m}_{ref}^H &\leq \dot{m}(\mathbf{z}) \leq 1.02 \cdot \dot{m}_{ref}^H, \\ r_{\min}(\mathbf{z}) &\geq r_{\min,ref} \end{aligned} \quad (20)$$

where, $\mathbf{z}$ is the latent variable of parameterized design space using GAN. More setting details of GAN-EGO will be illustrated in Section 4.3.

Table 2

Neuron network configuration of GAN for turbine blade.

Generator G	Discriminator D
Input $\in \mathbb{R}^{10}$	Input blade profile represented by 202 scatter points
FC 1024, BN, Leaky	Conv1D(2, 32, 3, 2, 1), BN, Leaky, Dropout
FC 256 $\times$ 12, BN, Leaky	Conv1D(32, 64, 3, 2, 1), BN, Leaky, Dropout
Deconv1D(256, 128, 5, 2, 1), BN, Leaky	Conv1D(64, 128, 3, 2, 1), BN, Leaky, Dropout
Deconv1D(128, 64, 5, 2, 1), BN, Leaky	Conv1D(128, 256, 3, 2, 1), BN, Leaky, Dropout
Deconv1D(64, 32, 5, 2, 2), BN, Leaky	Conv1D(256, 512, 3, 2, 1), BN, Leaky, Dropout
Deconv1D(32, 2, 5, 2, 2), BN, Leaky	FC 1024, BN, Leaky
FC 50, Sigmoid, output for FFD variables	FC 1, output for D

The third type of design optimization framework is the generative transfer optimization (GTO) framework proposed in this paper. Under the GTO framework, the construction of the latent space with sample-weighted GAN is the same as that of GAN-EGO. However, different from GAN-EGO, the source samples (i.e., $\{\mathbf{x}_S, \mathbf{y}_S\}_m$) are encoded into the target design space using gradient-based parameterization method shown in Section 3.2. Then, transfer optimization is performed with the multi-fidelity surrogate shown in Section 3.3. The above method is denoted by GAN-TO, which is expressed as follows:

$$\begin{aligned} \min\{T|S, K(t)\} &= \min\{T(\mathbf{G}(\mathbf{z}))|\{\mathbf{x}_S, \mathbf{y}_S\}_m, K(t)\} \\ &= \min\{T(\mathbf{G}(\mathbf{z}))|\{\mathbf{z}_S, \mathbf{y}_S\}_m, K(t)\} \end{aligned} \quad (21)$$

where, $\{\mathbf{z}_S, \mathbf{y}_S\}_m$ is the variable vector of optimized solutions that have been encoded in the latent design space using the gradient-based parameterization method shown in Section 3.2, and $K(t)$ is the multi-fidelity surrogate based knowledge transfer strategy that introduced in Section 3.3. More details of GAN-TO will be illustrated in Section 4.3.

4.2. Settings and testing of sample-weighted GAN

It is noted that the blade parameterization using sample-weighted GAN is of vital importance that determines the effectiveness of proposed generative transfer optimization method. In particular, the ability to accurately affine source samples to the parameterized design space of target optimization problem is a prerequisite for efficient transfer optimization thereafter. Hence, before detailed discussion of the settings of FFD-EGO, GAN-EGO and GAN-TO, the settings of GAN is illustrated in this subsection. The accuracy of gradient-based reparameterization of source samples is also tested.

4.2.1. Data preparation and settings of sample-weighted GAN

As illustrated in Section 3.1, the training dataset, the neuron net architecture, the setting of the regularization term in the loss function and sample-weighted training strategy as well directly determine the effectiveness of the generative model based parameterization.

Regarding the training dataset, it is made up of two parts, i.e., the unlabeled samples X_U and the optimized solutions of the source task $\{\mathbf{x}_S, \mathbf{y}_S\}_m$. The samples in X_U are obtained through sampling with the modified Pritchard's geometric parameterization method, which takes the highly loaded blade shown in Table 1 as the reference design. The inlet geometric angle, upper wedge angle, outlet flow angle, deflection angle and angle of circle center line of the blade as well are taken as variables, which can be changed by $\pm 10\%$ w.r.t. the reference design. The Latin hypercube sampling is used to generate the samples in X_U , and 1030 blade profiles are generated after filtering the irregular distorted blade profiles. As for $\{\mathbf{x}_S, \mathbf{y}_S\}_m$, the solutions of Eq. (16) are sorted by their objective function, and the top 40 solutions are selected as the optimized solutions of source task in $\{\mathbf{x}_S, \mathbf{y}_S\}_m$.

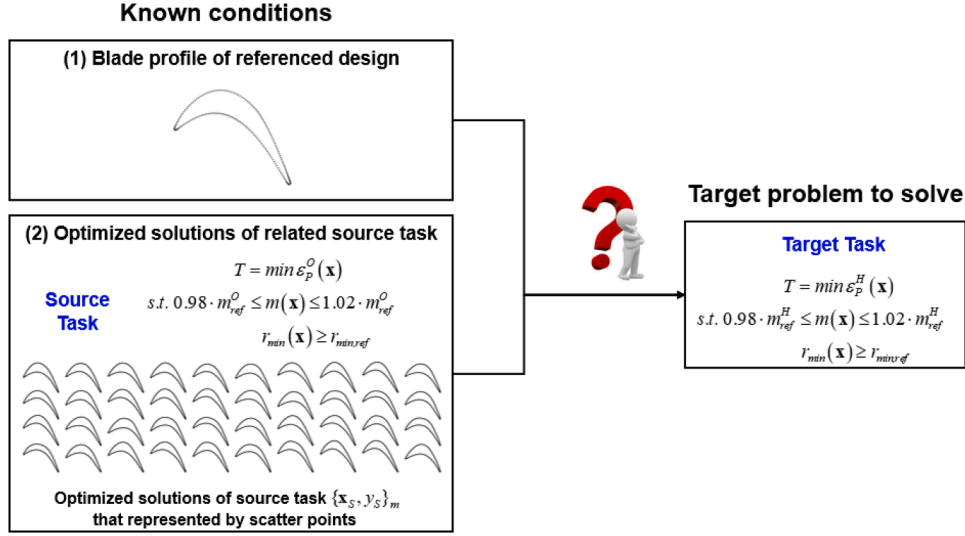


Fig. 9. Problem description of the optimization task for a highly loaded turbine blade.

The configuration of the neuron network architecture of GAN is shown in Table 2. Both the fully-connected neuron layer and one-dimensional convolution neuron layer are used. Similar to the work of Chen et al. [20], a FFD layer is imposed as the output layer of the generator in order to ensure the smoothness of generated blade profiles. Concerning the regularization term of the loss function L_p in Eq. (10) of Section 3.1, the following regularization term is used:

$$L_p = \mathbb{E}_{\mathbf{x}} \left(\max \kappa_{\max} \right), \kappa_{\max} = \max \frac{|\ddot{x}\ddot{y} - \ddot{x}\ddot{y}|}{[\ddot{x}^2 + \ddot{y}^2]^{1.5}} \quad (22)$$

where, $\ddot{x}$, $\ddot{y}$ are calculated using central difference, and $\kappa_{\max}$ is the maximum curvature of the blade profile. By minimizing the maximum curvature of the curve, the abnormal blade profiles that include abrupt corners and crossed curves can be avoided.

Considering the fact that the number of samples in X_U is much larger than that of $\{\mathbf{x}_S, \mathbf{y}_S\}_m$ in the training dataset, the sample-weighted training scheme is used in order to learn the distinct features of the optimized solutions of the source task. Specifically, the sample ratio in a batch is set to be 1, i.e., half of the samples in a batch of each epoch are randomly drawn from X_U and $\{\mathbf{x}_S, \mathbf{y}_S\}_m$, respectively. Moreover, adaptive learning rate is used for the training process. In particular, the total number of training epochs is set to be 4000, and the learning rate is fixed at 0.0001 in the first 2000 epochs, and then it linearly reduces to 0 at the 4000th epoch. During the model training process, maximum mean discrepancy (MMD) [40] is applied for monitoring the generation capability of GAN and reducing the risk of mode collapse. Additionally, after sensitivity analysis on latent dimension, the number of latent variables of GAN is set to 5, i.e., the dimension of vector $\mathbf{z}$ is 5.

4.2.2. Testing of gradient-based reparameterization accuracy

To examine the reparameterization accuracy of source samples using the gradient-based method shown in Section 3.2, the following testing procedure is conducted. That is, the sample $\mathbf{x}$ that represented by coordinate-based scatter points is first parameterized using the gradient-based reparameterization method illustrated in Section 3.2, which is labeled $\hat{\mathbf{x}}$. Then, with $\mathbf{z}$ as the input, the sample generator $G(\mathbf{z})$ is used to reconstruct the sample profile that represented by scatter points, denoted by $\hat{\mathbf{x}}$. By comparing $\mathbf{x}$ and $\hat{\mathbf{x}}$, the effectiveness of gradient-based reparameterization method can be examined. Following this way, the original and reparameterized samples are shown in Fig. 11, which are represented by solid and dashed lines, respectively. The solid and dashed lines are perfectly overlapped, indicating that the proposed gradient-based reparameterization method of source samples is accurate for transfer optimization.

4.3. Settings of compared design optimization methods

According to the references [41,42], the number of variables for FFD-based blade parameterization is set to 12. The number of variables for the GAN model is set to 5, after sensitivity analysis on the variable dimension, and the sensitivity analysis will be discussed with more details in the Section 4.5. In the meantime, to show the influence of the multi-fidelity surrogate modeling on the convergence rate of transfer optimization, both the most popularly used co-kriging-based MFS [43–45] and the EMFS shown in Section 3.3 are employed for the GAN-TO, labeled as GAN-TO-MFS and GAN-TO-EMFS, respectively. For fair comparison, the Latin hypercube sampling is used in FFD-EGO, GAN-EGO, GAN-TO-MFS and GAN-TO-EMFS to generate the initial samples, and the initial sample size is set to 40. Moreover, the total number of sample evaluations for each optimization method is controlled to be less than 140 samples per run. Additionally, to avoid the influence of random sampling on the optimization results, each optimization method will run 10 times and the averaged results will be taken for comparison.

4.4. Results discussion

The optimization results of FFD-EGO, GAN-EGO, GAN-TO-MFS and GAN-TO-EMFS over 10 runs are shown in Fig. 12. Wherein, Fig. 12(a) presents the mean convergence rate of the three compared design optimization methods, and Fig. 12(b) shows the statistical distributions of the final solutions, the middle line of each boxplot shows the medians of optimal solutions of FFD-EGO, GAN-EGO, GAN-TO-MFS and GAN-TO-EMFS, respectively.

Since the parameterized design space built by FFD cannot be the same as that constructed using GAN, the starting point of the FFD-EGO method is not overlapped with those of GAN-EGO, GAN-TO-MFS and GAN-TO-EMFS. The convergence rate of FFD-EGO is slower, and the performance of the final solution of FFD-EGO is worse, as shown in Fig. 12(a) and (b). The above conclusion is similar to the optimization results of Li et al. [16]. The possible reasons for this are as follows, compared to the GAN parameterized model obtained by training based on thousands of groups of type line samples, the sample diversity of the FFD that built based on a single reference design is less sufficient. In particular, considering the interactions between variables in parameterized space of FFD, the designers usually control the variation range of parameterized variables to reduce the ratio of abnormal profiles and to ensure that the optimization can be carried out smoothly; this further weakens the sample diversity of FFD-based parameterization space. In contrast,

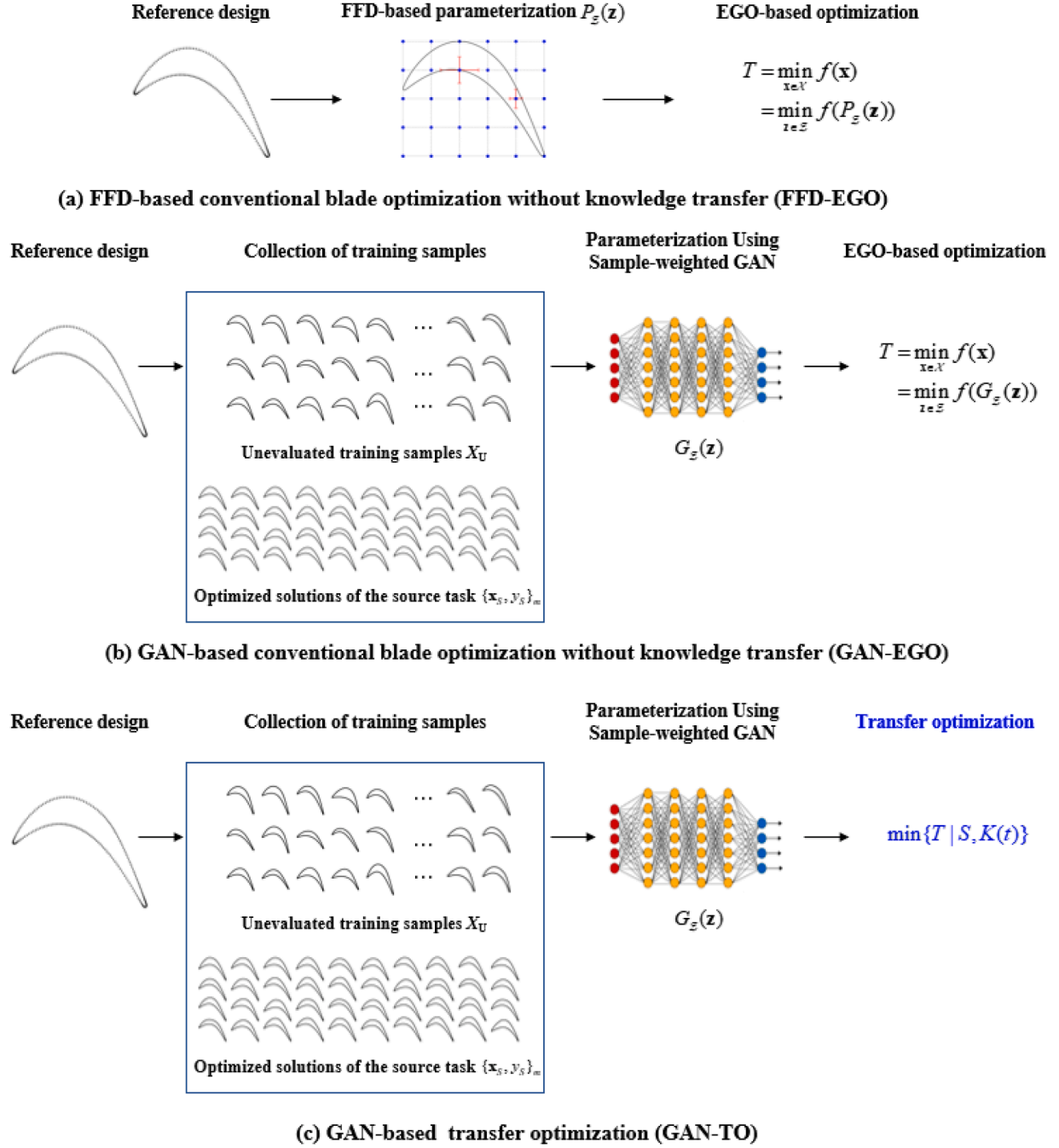


Fig. 10. Alternative turbine design methods for comparison.

by intelligently extracting geometric features from thousands of profile samples with deep learning model, the learned latent design space of GAN has a low rate of distortion profiles, the diversity of samples in related design space is also much better. Moreover, the inclusion of source samples (i.e., $\{\mathbf{x}_S, y_S\}_m$) as part of the training dataset and using the sample-weighted scheme to learn the distinct features of $\{\mathbf{x}_S, y_S\}_m$, the proportion of solutions with better performance in the target design space can be greatly increased. Thereby, GAN-EGO, GAN-TO-MFS and GAN-TO-EMFS can achieve better final solutions with faster convergence rates when comparing to those of FFD-EGO.

When comparing GAN-EGO with GAN-TO methods, the starting point of them are the same. However, the convergence rates of both GAN-TO-MFS and GAN-TO-EMFS are faster than that of GAN-EGO, and the performance of final solutions obtained by GAN-TO-MFS and GAN-

TO-EMFS are also look better than those of GAN-EGO. In the meantime, when the goal is to achieve similar optimization results as that achieved by GAN-EGO, the function calls that needed by GAN-TO can be reduced by 50% and even more when comparing that of GAN-EGO. Additionally, as shown in Fig. 10(b), the performance of both GAN-TO-MFS and GAN-TO-EMFS are more robust than that of GAN-EGO, as the variation range of the optimal solutions of both GAN-TO-MFS and GAN-TO-EMFS are much smaller than that of GAN-EGO. Further, when the two versions of GAN-TO are compared, the GAN-TO-EMFS is shown to have even faster convergence rate than that of GAN-TO-MFS. The reason behind can be explained by referring to the multi-fidelity modeling of the toy problem shown in Fig. 6. That is, while the MFS helps to capture the overall trend of the design space, the local SFS can better capture the local trend of the promising areas where HF samples with relatively better perfor-

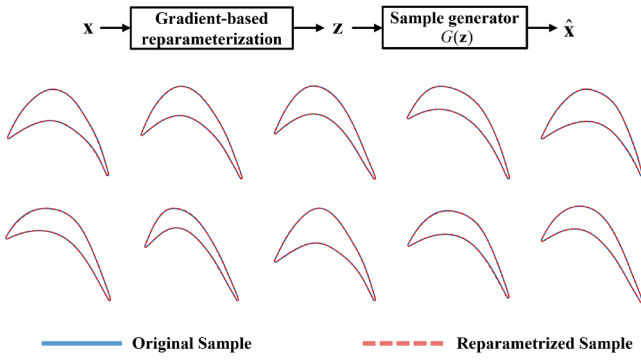
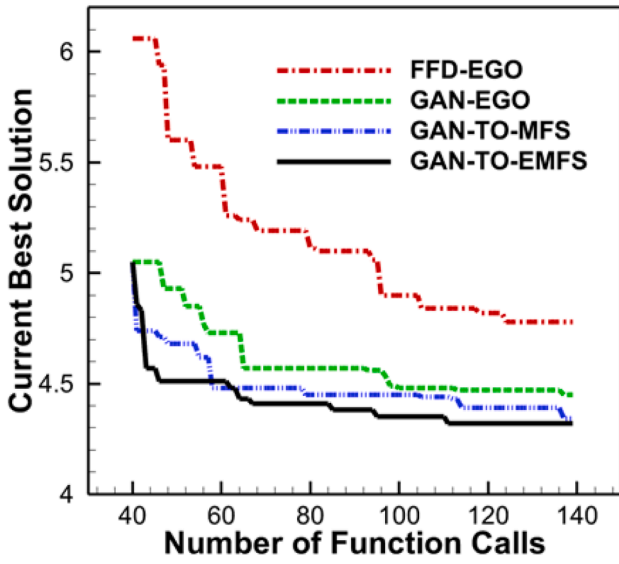
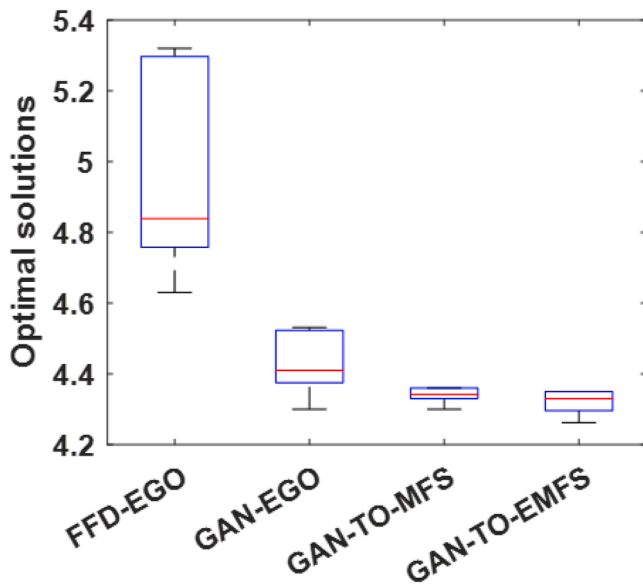


Fig. 11. Comparison of the blade profiles of the original and reparametrized samples.



(a) Mean convergence rate



(b) Distributions of optimal solutions

Fig. 12. Optimization performance of different design methods.

Table 3

Comparison of the reference and optimized designs.

	$\epsilon_p / \%$	$\dot{m} / \text{kg}/(\text{s} \cdot \text{m})$	Outlet flow angle/ $^\circ$
Reference	12.99	475.02	-60.86
FFD-EGO	4.72	471.50	-61.30
GAN-EGO	4.56	474.08	-61.05
GAN-TO-MFS	4.35	473.64	-61.10
GAN-TO-EMFS	4.29	467.70	-61.69

mance are more densely distributed. Thereby, as a combination of MFS and SFS, the EMFS can more accurately guide the algorithm to arrive the vicinity of the true optimal, leading to faster convergence rate with even better final solutions.

To further examine the performance of FFD-EGO, GAN-EGO, GAN-TO-MFS and GAN-TO-EMFS, the related representative solutions are analyzed. Table 3 shows the overall performance of the compared solutions. The total pressure loss of FFD-EGO, GAN-EGO and GAN-TO-MFS and GAN-TO-EMFS are all greatly reduced when compared to the reference design. In the meantime, the mass flow rate and outlet flow angle of optimized solutions meet the constraint shown in Eq. (18).

Fig. 13 shows the streamlines and velocity contours of the compared designs, and Fig. 14 presents the pressure distributions at the blade surface. Compared to the reference design shown in Fig. 13(a), the separation bubble near the blade trailing edge becomes almost invisible for the optimized solutions of FFD-EGO, GAN-EGO, GAN-TO-MFS and GAN-TO-EMFS. Correspondingly, as can be seen in Fig. 14, the flat area near the trailing edge of blade suction surface also disappeared for the solutions of FFD-EGO, GAN-EGO, GAN-TO-MFS and GAN-TO-EMFS. It indicates that the loss near the blade trailing edge of the optimized solutions are all reduced. Compared to the FFD-EGO design, the blade loadings of GAN-EGO is more forward, but the adverse pressure gradient near the trailing edge seems smoother. Thereby, the total pressure loss coefficient of GAN-EGO is smaller than that of FFD-EGO. Additionally, second accelerations are observed for both GAN-TO-MFS and GAN-TO-EMFS, which is of great help to reduce the pressure gradient along the suction surface, particularly the adverse pressure gradients. As a consequence, the designs of GAN-TO-MFS and GAN-TO-EMFS are shown to achieve even smaller total pressure loss coefficient. With the above, the effectiveness of the proposed generative transfer optimization method is demonstrated.

4.5. Reasoning on the effectiveness of transfer optimization and sensitivity analysis

To examine the reason why transfer optimization can finish the task more quickly and achieving better solutions, the correlation between the target optimal solutions and the source samples are analyzed, as shown in Figs. 15 and 16. In Fig. 15, the closeness of the solutions of the source and target problems is analyzed by the t-SNE plot. Specifically, the horizontal and vertical coordinates of the scatter points in the t-SNE plot show the closeness of samples. That is, when the samples are located close to each other, their geometric features are similar and vice versa. As shown in Fig. 15, the optimized solutions of the source task and the optimized solutions of GAN-TO and those of GAN-EGO as well are all located in the upper left region, indicating that the solution features of the source and target problem are similar. Therefore, when training the latent design space of the target problem using sample-weighted GAN, the solutions of the source task can help to increase the proportion of high-performance samples in the target design space.

As another evidence, Fig. 16 shows the correlation of the objective function between the source task target task. A set of blade profiles are generated using GAN, which are evaluated under the settings of the target and source problem, respectively. The objective

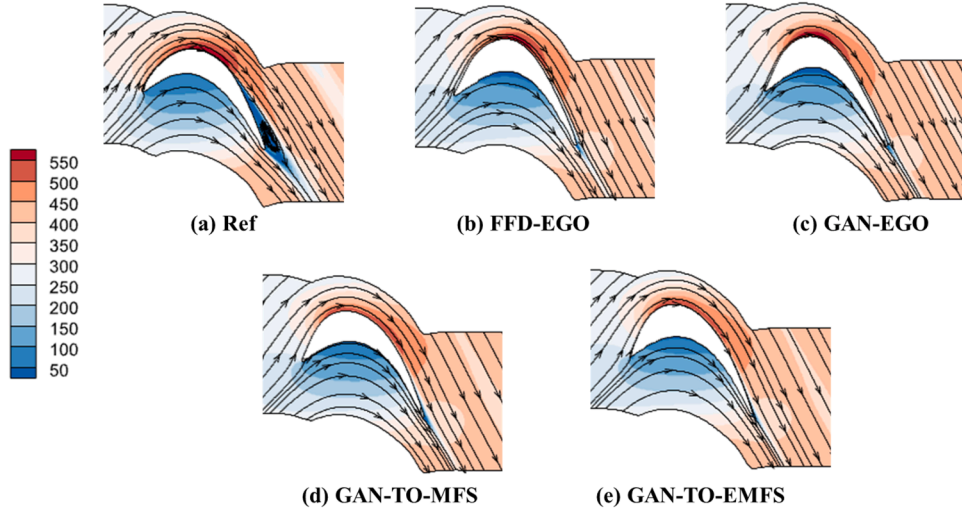


Fig. 13. Streamlines and velocity contours of the compared designs.

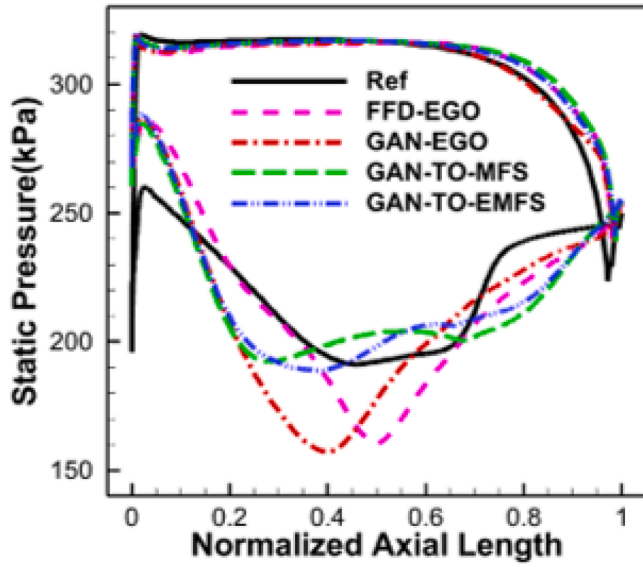


Fig. 14. Pressure distribution at the blade surface of the compared designs.

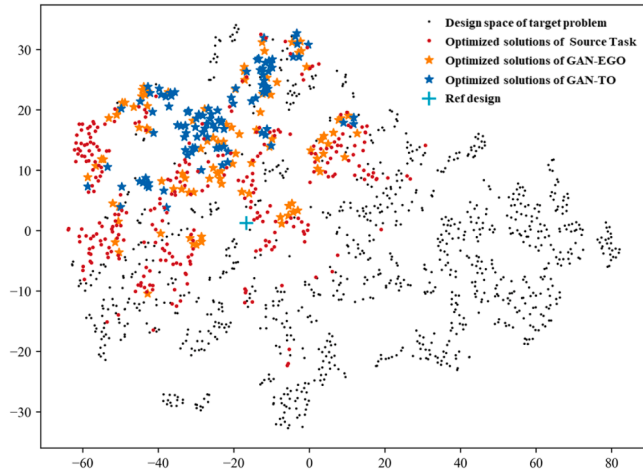


Fig. 15. Sample distributions of the source and target tasks.

Table 4

Sensitivity analysis of latent dimension of GAN on the reparameterization accuracy.

Latent dimension	Reconstruction error measured by the normalized MSE)
$d = 3$	$(1.886 \pm 1.072) \times 10^{-5}$
$d = 5$	$(1.207 \pm 0.538) \times 10^{-5}$
$d = 13$	$(1.013 \pm 0.432) \times 10^{-5}$
$d = 30$	$(1.005 \pm 0.385) \times 10^{-5}$

function values of the target task and source task are plotted, as shown in Fig. 16, and their correlation is calculated using the Pearson's correlation coefficient, i.e., $r = 0.9037$. It indicates that the objectives of the target and source task are highly correlated. Thereby, the multi-fidelity surrogate that used in GAN-TO can draw useful information from the source samples to improve the approximation accuracy of the target objective function, which further helps the algorithm to arrive at the vicinity of the optimal solution of target task more quickly.

Furthermore, to show the influence of latent dimension of GAN on the transfer optimization, the sample-weighted GAN is trained with different latent dimension. Specifically, the settings and the neuron network architecture of different GAN models remain the same as that shown in Section 4.2. The differences between the GAN models is that, their input dimensions are different. Table 4 shows the influence of the latent dimension on the blade reparameterization accuracy, where the same testing set of blade profiles shown in Fig. 11 are used for the test. Fig. 17 shows the influence of latent dimension on the optimization convergence rates over 10 runs, where the optimization settings keeps the same as those introduced in Section 4.4. It shows that, with the increase of latent dimension, the blade reparameterization accuracy increases; however, there exists an optimal latent dimension which can achieve the best final solution with faster convergence within budget. The reason behind can be explained as follows. With the increase of latent dimension, the GAN can better capture the sophisticated local geometric features of blade profiles, and thereby the reparameterization accuracy increase. It also means that, the performance of true optimal solution of the design space is better with the increase of latent dimension. However, with the increase of latent dimension, more training samples are required to build an accurate surrogate that can well capture the function trend of the design space. In other words, there should be a trade of between the computational cost and the blade performance in the optimization search. Then, with the same number of initial samples and the same

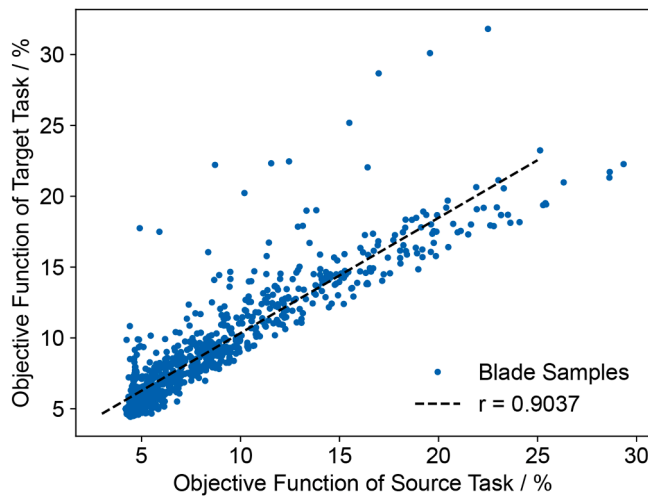


Fig. 16. Correlation of objective functions of the source and target tasks in turbine blade optimization.

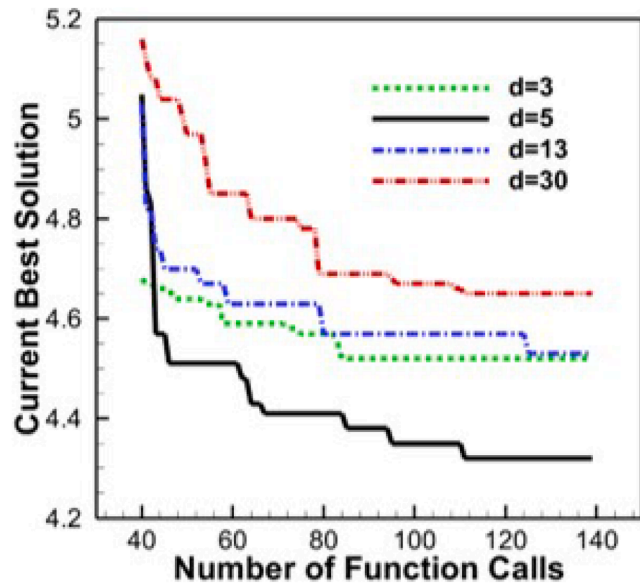


Fig. 17. Sensitivity analysis of the latent dimension of GAN on the optimization convergence rate.

sample budget as well, the optimization with $d = 3$ can converge most quickly but with less optimized solutions, while the optimizations with $d = 30$ are observed to attain the solutions with worst performance. As a tradeoff, the optimizations with $d = 5$ achieves the best performance in such settings.

5. Conclusions

To speed up turbine blade design, a generative transfer optimization framework is proposed. Some concluding remarks are drawn as follows.

(1) A sample-weighted scheme is proposed for the GAN training, which includes the solutions of related tasks as part of the training data to increase the proportion of high-performance samples in the learned latent space for the target optimization problem. Moreover, a gradient-based reparameterization method is proposed to overcome the weakness of GAN that cannot directly conduct inverse inference, and thus the source samples represented by coordinate points can be accurately encoded into the latent space of the target task, allowing the use of source samples for efficient transfer optimization.

(2) A multi-fidelity surrogate-based transfer optimization strategy is proposed. Then, combining it with the sample-weighted GAN and gradient-based reparameterization method, a generative transfer optimization framework is proposed for more efficient turbine design. Through transfer optimization of a turbine blade with high loadings and compared it to the conventional methods over 10 runs, the proposed transfer optimization method is observed to finish the target task more quickly and achieving better solutions. When the goal is to attain solutions with similar performance, the number of performance evaluations required by the proposed transfer optimization framework can be reduced by 50% and even more.

(3) Through sensitivity analysis of the influence of latent dimension on the optimization convergence rate, it shows that with the increase of latent dimension, the blade reparameterization accuracy increases; however, there exists an optimal latent dimension to achieve the best final solution within budget.

Also note that, the proposed generative transfer optimization framework can be easily extended to accelerate the optimization progress of other turbomachinery applications. In the meantime, to achieve the better solutions more quickly, more efforts should be made on how to efficiently find the optimal latent dimension of the generative model.

CRedit authorship contribution statement

Fei Zeng: Investigation; **Cunxi Li:** Methodology; **Liming Song:** Conceptualization; **Zhendong Guo:** Writing – original draft, Methodology, Investigation.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary material

Supplementary material associated with this article can be found in the online version at [10.1016/j.ast.2026.112324](https://doi.org/10.1016/j.ast.2026.112324)

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